

● MEMBER 1 – Course & Content Management

1 Courses Table

Field Name	Data Type	Description
_id	ObjectId (PK)	Unique course ID
title	String	Course name
description	String	Course details
category	String	Course category
subject	String	Subject area
thumbnailUrl	String	Course image
createdBy	ObjectId (FK → Users)	Tutor/Admin who created
isPremium	Boolean	Free or paid course
price	Number	Price if premium
createdAt	Date	Creation date
updatedAt	Date	Last update

2 LectureMaterials Table

Field Name	Data Type	Description
_id	ObjectId (PK)	Unique material ID
courseId	ObjectId (FK → Courses)	Related course
title	String	Material title
description	String	Material description
week	Number	Week number
topic	String	Topic name
fileUrl	String	File storage link

Field Name	Data Type	Description
fileType	String	pdf/video/doc
uploadedBy	ObjectId (FK → Users)	Uploaded by
createdAt	Date	Upload date

3 Enrollments Table

Field Name	Data Type	Description
_id	ObjectId (PK)	Enrollment ID
userId	ObjectId (FK → Users)	Student
courseId	ObjectId (FK → Courses)	Course
enrolledAt	Date	Enrollment date
paymentStatus	String	Free/Paid

MEMBER 3 – User Management & Kuppi Booking

4 Users Table

Field Name	Data Type	Description
_id	ObjectId (PK)	User ID
name	String	Full name
email	String (Unique)	Login email
password	String	Hashed password
role	String	Student/Tutor/Admin
profileImage	String	Profile picture URL
bio	String	Short description
subjectsOfInterest	Array	Interested subjects

Field Name	Data Type	Description
createdAt	Date	Registration date

5 KuppiSessions Table

Field Name	Data Type	Description
_id	ObjectId (PK)	Session ID
tutorId	ObjectId (FK → Users)	Tutor
subject	String	Subject
description	String	Session details
date	Date	Session date
startTime	String	Start time
endTime	String	End time
duration	Number	Minutes
maxParticipants	Number	Capacity
isPremium	Boolean	Paid session
price	Number	Session price
createdAt	Date	Created date

6 SessionBookings Table

Field Name	Data Type	Description
_id	ObjectId (PK)	Booking ID
sessionId	ObjectId (FK → KuppiSessions)	Session
studentId	ObjectId (FK → Users)	Student
bookingStatus	String	Confirmed/Cancelled
paymentStatus	String	Paid/Unpaid

Field Name	Data Type	Description
bookedAt	Date	Booking time

MEMBER 2 – Quiz & Mock Exam System

Quizzes Table

Field Name	Data Type	Description
_id	ObjectId (PK)	Quiz ID
courseId	ObjectId (FK → Courses)	Related course
title	String	Quiz title
description	String	Quiz description
difficulty	String	Easy/Medium/Hard
timeLimit	Number	Minutes
totalMarks	Number	Total marks
isPremium	Boolean	Paid quiz
createdBy	ObjectId (FK → Users)	Tutor/Admin
createdAt	Date	Creation date

Questions Table

Field Name	Data Type	Description
_id	ObjectId (PK)	Question ID
quizId	ObjectId (FK → Quizzes)	Related quiz
questionText	String	Question
questionType	String	MCQ/TF/ShortAnswer
options	Array	MCQ options

Field Name	Data Type	Description
correctAnswer	String	Correct answer
marks	Number	Marks for question

9 QuizAttempts Table

Field Name	Data Type	Description
_id	ObjectId (PK)	Attempt ID
quizId	ObjectId (FK → Quizzes)	Quiz
userId	ObjectId (FK → Users)	Student
answers	Array	Selected answers
score	Number	Final score
percentage	Number	Percentage
startedAt	Date	Start time
submittedAt	Date	Submission time

10 ProgressAnalytics Table (Optional)

Field Name	Data Type	Description
_id	ObjectId (PK)	Analytics ID
userId	ObjectId (FK → Users)	Student
courseId	ObjectId (FK → Courses)	Course
quizzesAttempted	Number	Total quizzes
averageScore	Number	Avg score
strengths	Array	Strong topics
weaknesses	Array	Weak topics
lastUpdated	Date	Last update

MEMBER 4 – AI Chatbot & Payment System

1 1 ChatHistory Table

Field Name	Data Type	Description
_id	ObjectId (PK)	Chat ID
userId	ObjectId (FK → Users)	Student
messages	Array	Chat messages
contextCourseId	ObjectId (FK → Courses)	Related course
createdAt	Date	Chat start date

1 2 Transactions Table

Field Name	Data Type	Description
_id	ObjectId (PK)	Transaction ID
userId	ObjectId (FK → Users)	Student
amount	Number	Payment amount
currency	String	LKR/USD
paymentGateway	String	Stripe/PayPal
paymentIntentId	String	Gateway ID
status	String	Pending/Success/Failed
purchasedItemType	String	Course/Quiz/Session
purchasedItemId	ObjectId	Related item
createdAt	Date	Transaction date

1 3 PurchasedContent Table

Field Name	Data Type	Description
_id	ObjectId (PK)	Purchase ID
userId	ObjectId (FK → Users)	Buyer
itemType	String	Course/Quiz/Session
itemId	ObjectId	Purchased item
purchasedAt	Date	Purchase date